

Davide Boscaini, Ph.D.

Teaching machines how to perceive and understand the surrounding environment

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About

I am a Research Scientist at the **Technologies of Vision** research unit of the **Fondazione Bruno Kessler** in Trento, Italy. My interests are in 3D perception and understanding, with a focus on object 6D pose estimation and 3D scene segmentation. According to Google Scholar, I have an h-index of 17, an i10-index of 24, and my articles have over 6,800 citations.

Before joining FBK, I received a PhD in Computational Science from the **Università della Svizzera italiana**

in Lugano, Switzerland, in 2017. During my PhD, under the supervision of prof. **Michael Bronstein**, my research focused on extending deep learning techniques to geometric domains such as 3D shapes and graphs, contributing to the birth of a new research direction called **Geometric Deep Learning**. Prior to that, I obtained an M.S. in Mathematics from the **University of Verona**, Italy, in 2013, and a B.S. in Applied Mathematics from the same institution in 2010.

Education

Ph.D. in Computational Science Università della Svizzera italiana Dissertation on “Geometric Deep Learning for Shape Analysis” Advisor: M.M. Bronstein. Co-advisor: J. Masci Examiners: J. Schmidhuber, M. Ovsjanikov, P. Vanderghenst, K. Hormann	Sep. 2013 – Sep. 2017 <i>Lugano, Switzerland</i>
M.S. in Mathematics University of Verona Dissertation on “Spectral Methods for Shape Analysis” Advisor: G. Orlandi. Co-advisor: U. Castellani	Oct. 2010 – Mar. 2013 <i>Verona, Italy</i>
B.S. in Applied Mathematics University of Verona Dissertation on “Existence and multiplicity of the solutions of the Plateau problem” Advisor: S. Baldo	Sep. 2007 – Oct. 2010 <i>Verona, Italy</i>

Awards

“Best overall method” for 6D detection of unseen objects	BOP Challenge 2024
“Best overall method” for 6D localization of unseen objects	BOP Challenge 2024
“Early-bird winner” of 6D detection of unseen objects	BOP Challenge 2024
“Early-bird winner” of 6D localization of unseen objects	BOP Challenge 2024
“Best method on TUD-L dataset” for the 6D localization of unseen objects	BOP Challenge 2023

Publications

Foundational feature fusion for conditional flow matching in 6D pose estimation A. Hamza, D. Boscaini, F. Poiesi <i>Flow matching can estimate 6D poses of seen objects from frozen foundational features</i>	BMVC 2026
Generative 6D pose estimation via conditional flow matching A. Hamza, D. Boscaini, W. Li, B. Busam, F. Poiesi <i>First application of conditional flow matching to 6D pose estimation of seen objects</i>	ICIP 2026
Action-guided generation of 3D functionality segmentation data J. Corsetti, F. Giuliari, D. Boscaini, P. Hermosilla, A. Pilzer, G. Mei, A. Delitzas, F. Engelmann, F. Poiesi <i>We generate functionality segmentation data as real-world ones are too scarce for training neural networks</i>	CVPR-W 2026
High-resolution open-vocabulary object 6D pose estimation J. Corsetti, D. Boscaini, F. Giuliari, C. Oh, A. Cavallaro, F. Poiesi	TPAMI, 2026
CHIP: A multi-sensor dataset for 6D pose estimation of chairs in industrial settings M. Nardon, M. Mujika, A. González, D. Sedano, J. Rueda, A. Caro, A. Caraffa, F. Poiesi, P.I. Chippendale, D. Boscaini	BMVC 2025

- Distilling 3D distinctive local descriptors for 6D pose estimation** IROS 2025
A. Hamza, A. Caraffa, D. Boscaini, F. Poiesi
- AI-driven visual monitoring of industrial assembly tasks** ICIAP 2025
M. Nardon, S. Messelodi, A. Granata, F. Poiesi, A. Danese, D. Boscaini
- Functionality understanding and segmentation in 3D scenes** CVPR 2025
J. Corsetti, F. Giuliari, A. Fasoli, D. Boscaini, F. Poiesi
Highlight poster (top 3%). Awarded "Second place" for "Task-driven affordance grounding" at the SceneFun3D Challenge 2025
- 3D part segmentation via geometric aggregation of 2D visual features** WACV 2025
M. Garosi, R. Tedoldi, D. Boscaini, M. Mancini, N. Sebe, F. Poiesi
- Enabling seamless HRC integration: The AI-PRISM reference architecture** I-ESA 2024
M.A. Mateo-Casalí, J. Rueda, S.O. Afolaranmi, J.L.M. Lastra, D. Boscaini, L. Moya-Ruiz, F. Fraile
- Towards more effective human-robot collaboration via accurate pose estimation** ERF 2024
M.A. Mateo-Casalí, L. Moya-Ruiz, A. Caraffa, D. Boscaini, A. Hamza, P. Chippendale, F. Poiesi, F. Fraile
- Wild Berry image dataset collected in Finnish forests and peatlands using drones** ECCV-W 2024
L. Riz, S. Povoli, A. Caraffa, D. Boscaini, M.L. Mekhalfi, P. Chippendale, M. Turtiainen, B. Partanen, L.S. Ballester, F.B. Noguera, A. Franchi, E. Castelli, G. Piccinini, L. Marchesotti, M.S. Couceiro, F. Poiesi
- FreeZe: Training-free zero-shot 6D pose estimation with geometric and vision foundation models** ECCV 2024
A. Caraffa, D. Boscaini, A. Hamza, F. Poiesi
An enhanced version of this work, FreeZe-v2, is the "early bird winner" in both 6D localization and 6D detection at the BOP Challenge 2024. An early version of this work, PoZe, won the "Best method on TUD-L dataset" award at the BOP Challenge 2023
- Exploring fine-grained retail product discrimination with zero-shot object classification using Vision-Language Models** RTSI 2024
A. Tur, A. Conti, C. Beyan, D. Boscaini, R. Larcher, S. Messelodi, F. Poiesi, E. Ricci
- Open-vocabulary object 6D pose estimation** CVPR 2024
J. Corsetti, D. Boscaini, C. Oh, A. Cavallaro, F. Poiesi
- Tracciamento 3D della palla da punti di vista multipli nella pallavolo** Ital-IA 2024
L. Riz, S. Povoli, D. Boscaini, S. Messelodi, F. Poiesi
- Detect, Augment, Compose, and Adapt: Four steps for unsupervised domain adaptation in object detection** BMVC 2023
M.L. Mekhalfi, D. Boscaini, F. Poiesi
- Revisiting Fully Convolutional Geometric Features for object 6D pose estimation** ICCV-W 2023
J. Corsetti, D. Boscaini, F. Poiesi
- PatchMixer: Rethinking network design to boost generalization for 3D point cloud understanding** IMAVIS, 2023
D. Boscaini, F. Poiesi
Novel network design that is intrinsically effective in generalisation across datasets unseen at training time
- Supervised tractogram filtering using geometric deep learning** MIA, 2023
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, S. Sarubbo, J. Masci, D. Boscaini, P. Avesani
- The MONET dataset: Multimodal drone thermal dataset recorded in rural scenarios** CVPR-W 2023
L. Riz, A. Caraffa, M. Bortolon, M.L. Mekhalfi, D. Boscaini, A. Moura, J. Antunes, A. Dias, H. Silva, A. Leonidou, C. Constantinides, C. Keleshis, D. Abate, F. Poiesi
- Learning general and distinctive 3D local deep descriptors for point cloud registration** TPAMI, 2023
F. Poiesi, D. Boscaini
State-of-the-art performance for point cloud registration in the transfer learning setting across 3DMatch, ETH, and Kitti datasets

- SHIELD: Safeguard heritage in endangered looted districts** Ital-IA 2022
M.L. Mekhalfi, N. Saljoughi, D. Boscaini, F. Poiesi
- Localisation of defects in volumetric CT scans of valuable wood logs** ICPR-W 2020
D. Boscaini, F. Poiesi, S. Messelodi, A. Younes, D. Grande
Selected for oral presentation
- Joint supervised and self-supervised learning for 3D real-world challenges** ICPR 2020
A. Alliegro, D. Boscaini, T. Tommasi
Selected for oral presentation (4.4% acceptance rate)
- Distinctive 3D local deep descriptors** ICPR 2020
F. Poiesi, D. Boscaini
- Shape consistent 2D keypoint estimation under domain shift** ICPR 2020
L.O. Vasconcelos, M. Mancini, D. Boscaini, S. Rota Bulò, B. Caputo, E. Ricci
- Novel-view human action synthesis** ACCV 2020
M. Lakhal, D. Boscaini, F. Poiesi, O. Lanz, A. Cavallaro
- Clustered dynamic graph CNN for biometric 3D hand shape recognition** IJCB 2020
J. Svoboda, P. Astolfi, D. Boscaini, J. Masci, M.M. Bronstein
- Tractogram filtering of anatomically non-plausible fibers with geometric deep learning** MICCAI 2020
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, J. Masci, D. Boscaini, P. Avesani
- Self-supervision for 3D real-world challenges** ECCV-W 2020
A. Alliegro, D. Boscaini, T. Tommasi
- Deciphering interaction fingerprints from protein molecular surfaces** Nature Methods, 2020
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira
Advertised on the cover of the Feb 2020 issue of the journal
- Learning interaction patterns from surface representations of protein structure** NeurIPS-W 2019
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira
- Structured domain adaptation for 3D keypoint estimation** 3DV 2019
L.O. Vasconcelos, M. Mancini, D. Boscaini, B. Caputo, E. Ricci
Oral presentation
- 3D shape segmentation with geometric deep learning** ICIAP 2019
D. Boscaini, F. Poiesi
Spotlight presentation
- Geometric deep learning on graphs and manifolds using mixture model CNNs** CVPR 2017
F. Monti*, D. Boscaini*, J. Masci, E. Rodolà, J. Svoboda, M.M. Bronstein
Oral presentation (top 0.8%). First unified framework generalizing CNN architectures to non-Euclidean domains such as 3D shapes and graphs. Also available as technical report: arXiv:1611.08402. (indicates equal contribution)*
- Geometric deep learning** SIGGRAPH Asia Courses 2016
J. Masci, E. Rodolà, D. Boscaini, M.M. Bronstein, H. Li
- Learning shape correspondence with anisotropic convolutional neural networks** NeurIPS 2016
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein
Presented also as a poster at the 3D Deep Learning Workshop (3DLL) 2016. Also available as technical report: arXiv:1605.06437
- Anisotropic diffusion descriptors** CGF, 2016
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein, D. Cremers
Oral presentation at EUROGRAPHICS 2016
- Geodesic convolutional neural networks on Riemannian manifolds** ICCV-W 2015
J. Masci*, D. Boscaini*, M.M. Bronstein, P. Vandergheynst
Oral presentation at 3DRR 2015. First extension of the popular CNN paradigm to non-Euclidean domains. An early version of this work was published as the technical report arXiv:1501.06297 on January 2015. (indicates equal contribution)*

Learning class-specific descriptors for deformable shapes using localized spectral convolutional networks

CGF, 2015

D. Boscaini, J. Masci, S. Melzi, M.M. Bronstein, U. Castellani, P. Vanderghelynst
Oral presentation at SGP 2015

Shape-from-operator: Recovering shapes from intrinsic operators

CGF, 2015

D. Boscaini, D. Eynard, D. Kourounis, M.M. Bronstein
Oral presentation at EUROGRAPHICS 2015. First approach able to synthesize the extrinsic geometry of a shape from intrinsic information. An early version of this work was published as the technical report arXiv:1406.1925 on June 2014

Coulomb shapes: Using electrostatic forces for deformation-invariant shape representation

EUROGRAPHICS-W 2014

D. Boscaini, R. Girdziusas, M.M. Bronstein
Oral presentation at 3DOR 2014. Presented also as a poster at the International Computer Vision Summer School (ICVSS), 2014

A sparse coding approach for local-to-global 3D shape description

The Visual Computer, 2014

D. Boscaini, U. Castellani
Invited paper. Journal extension of the 3DOR 2013 conference paper

Local signatures quantization by sparse coding

EUROGRAPHICS-W 2013

D. Boscaini, U. Castellani
Oral presentation at 3DOR 2013. Presented also as a poster at SGP 2013

Patents

US patent application No. 17675011

Clustered dynamic graph convolutional neural network for biometric 3D hand recognition
Inventors: J. Svoboda, P. Astolfi, D. Boscaini, J. Masci

US patent No. 10210430

Filed Feb. 19, 2019

System and a method for learning features on geometric domains (CIP)
Inventors: M.M. Bronstein, D. Boscaini, F. Monti • Acquired by Twitter Inc.

US patent No. 10013653

Filed Jul. 3, 2018

System and a method for learning features on geometric domains
Inventors: M.M. Bronstein, D. Boscaini, J. Masci, P. Vanderghelynst • Acquired by Twitter Inc.

Invited talks

Object 6D pose estimation with vision and language

Dec. 18, 2025

University of Catania, Italy • Invited by Francesco Ragusa, Giovanni Farinella

3D object understanding on the shoulders of 2D foundation models

Jul. 29, 2025

Shandong University, China • Invited by Fabio Poiesi

Object 6D pose estimation in the foundation models era

Nov. 22, 2024

University of Trento, Italy • Master course in “Trends and Applications in Computer Vision”, Invited by Massimiliano Mancini

Object 6D pose estimation in the foundation models era

Jun. 6, 2024

Politecnico di Torino, Italy • Invited by Francesca Pistilli

3D object understanding on the shoulders of 2D foundation models

Mar. 28, 2024

École Polytechnique, Paris, France • Invited by Maks Ovsjanikov

3D deep learning to the test of real-world challenges

Dec. 11, 2020

University of Pisa, Italy • Ph.D. event “Visions of Tomorrow”

3D Deep Learning

Dec. 11, 2019

Politecnico di Torino, Italy • Invited by Tatiana Tommasi

Geometric deep learning for 3D shape analysis

May 13, 2019

Politecnico di Torino, Italy • Invited by Barbara Caputo

Geometric deep learning for shape analysis

Sep. 2, 2017

Kos, Greece • Keynote at EUSIPCO 2017

Geometric deep learning for shape analysis

Apr. 4, 2017

FBK-TeV, Trento, Italy • Invited by Samuel Rota Bulò and Stefano Messelodi

Geometric deep learning for shape analysis CNR-IMATI, Genoa, Italy • Invited by Michela Spagnuolo	Feb. 13, 2017
Deep learning on geometric data Zurich, Switzerland • SSSTC RiC big data research workshop	Feb. 16, 2016
Deep learning on geometric data Embedded Vision Systems (eVS), Verona, Italy • Invited by Roberto Marzotto	Feb. 8, 2016
Deep learning on geometric data Rainbow group, University of Cambridge, UK • Invited by Flora Tasse	Feb. 4, 2016
Deep learning on geometric data University of Cambridge, UK • C.A.K.E. seminar, Invited by Simone Parisotto	Feb. 3, 2016
Convolutional neural networks on non-Euclidean domains Potsdam, Germany • Keynote at SciCADE 2015	Sep. 14, 2015
Shape-from-operators: recovering shapes from intrinsic differential operators TUM, Munich, Germany • Invited by Emanuele Rodolà	Nov. 26, 2014
Shape-from-operators: recovering shapes from intrinsic differential operators Disentis, Switzerland • USI-ICS retreat	Aug. 19, 2014

Teaching experience

Academic courses

Trends and Applications in Computer Vision	University of Trento, Fall 2023
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Short courses and tutorials

Functional Maps: A Flexible Representation for Learning and Computing Correspondences	3DV 2018
Geometric Deep Learning	SIGGRAPH Asia 2016
Deep Learning for Shape Analysis	EUROGRAPHICS 2016

Teaching Assistantships

Computer Vision and Pattern Recognition	Università della Svizzera italiana, Spring 2017
Computer Vision and Pattern Recognition	Università della Svizzera italiana, Spring 2016
Large Scale Optimization	Università della Svizzera italiana, Spring 2016
Computer Graphics	Università della Svizzera italiana, Fall 2014
Geometric Image Processing and Computer Vision	Università della Svizzera italiana, Spring 2014
Calculus	Università della Svizzera italiana, Fall 2013
Mathematical Analysis 1, Mathematical Analysis 2	University of Verona, 2012–2013
Mathematical Analysis 1, Mathematical Analysis 2	University of Verona, 2011–2012
Mathematical Analysis 1	University of Verona, 2010–2011

Student supervision

Jaime Corsetti, PhD student at FBK and University of Trento Project: Vision-language models for embodied AI	Nov. 2023–present
Mattia Nardon, Master student at University of Trento Role: Internship and Master thesis advisor Project: AI-powered visual monitoring of Lego assembly tasks	Mar.–Dec. 2024
Alice Fasoli, Master student at University of Trento Role: Internship and Master thesis advisor Project: Retrieval-driven 6D pose estimation of unseen objects	Mar.–Dec. 2024
Matteo Minardi, Master student at University of Trento Role: Internship and Master thesis advisor Projects: Eye-gaze estimation using smart glasses, Vision encoder role in VLMs	Mar.–Oct. 2024
Marco Garosi, Master student at University of Trento Project: Zero-shot semantic segmentation of 3D objects Outcome: Publication at WACV 2025	Dec. 2023–Apr. 2024
Riccardo Tedoldi, Master student at University of Trento	Dec. 2023–Apr. 2024

Project: Zero-shot semantic segmentation of 3D objects	
Outcome: Publication at WACV 2025	
Jaime Corsetti, Master student at University of Trento	2022–Oct. 2023
Projects: Open-vocabulary and Supervised object 6D pose estimation for RGBD images	
Outcome: Publication at CVPR 2024	
Safa Abbes, Master student at University of Trento	2022–2023
Role: Masther thesis coadvisor	
Project: Self-supervised domain adaptation for RGB images	
Antonio Alliegro, PhD student at Politecnico di Torino	2020–2021
Project: Self-Supervised domain adaptation for 3D point clouds	
Outcome: Publications at ECCV-W 2020 and ICPR 2021	
Pietro Astolfi, PhD student at FBK, UniTN, and IIT	2019–2021
Project: Geometric Deep Learning for brain structure analysis	
Outcome: Publications at MICCAI 2020, IJCB 2020, and MIA 2023	
Levi O. Vasconcelos, PhD student at UniTN and IIT	2019–2020
Project: Structured domain adaptation	
Outcome: Publications at 3DV 2019 and ICPR 2020	
Antonio Alliegro, Master student at Politecnico di Torino	2019–2020
Role: Masther thesis coadvisor	
Piero Cavalcanti, Master student at Politecnico di Torino	2019–2020
Role: Masther thesis coadvisor	
Myriam Bronstein, Master student at Università della Svizzera italiana	2016
Project: Machine learning methods on manifolds and graphs	
Fatemeh Chegini, Master student at Università della Svizzera italiana	2014–2015
Project: Spectral methods for cross-modal retrieval	

Academic service

Conferences revision activity

International Conference on Robotics and Automation (ICRA)	2022, 2020
International Conference on Pattern Recognition (ICPR)	2022, 2020
Symmetry and Geometry in Neural Representations (NeurIPS Workshops)	2022
International Conference on Image Analysis and Processing (ICIAP)	2022
International Conference on Machine Learning, Optimization, and Data Science (LOD)	2022
Symposium On Applied Computing (SAC)	2022
International Conference on 3D Vision (3DV)	2021, 2020, 2019, 2018
International Conference on Machine Learning, Optimization, and Data Science (LOD)	2021
International Conference on Machine Vision Applications (MVA)	2021, 2019
EUROGRAPHICS	2019, 2017, 2015
The British Machine Vision Conference (BMVC)	2018
Computer Vision and Pattern Recognition (CVPR)	2017
International Symposium on Vision, Modeling and Visualization (VMV)	2016
Neural Information Processing Systems (NeurIPS)	2016

Journal revision activity

Robotics and Automation Letters (RAL)	2022
Computer Graphics Forum (CGF)	2022
IEEE Transactions on Image Processing (TIP)	2022, 2021
IEEE Transactions on Transactions on Knowledge and Data Engineering (TKDE)	2022, 2021
Neural Processing Letters (NEPL)	2022
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)	2021, 2020
IEEE Transactions on Visualization and Computer Graphics (TVCG)	2020, 2018, 2017
Computers and Graphics	2019

Computer Vision and Image Understanding (CVIU)	2019, 2015
International Journal of Machine Learning and Cybernetics (JMCL)	2019
Pattern Recognition Letters	2019
The Visual Computer Journal (TVCJ)	2018, 2017, 2016
Computer Aided Geometric Design (CAGD)	2018
Computer-Aided Design (CAD)	2018
Sensors	2018
IPSJ Transactions on Computer Vision and Applications	2017

Area chair

British Machine Vision Conference (BMVC)	2025
British Machine Vision Conference (BMVC)	2024

Program committee

International Workshop on Advances in Drone Vision (ADV)	2025
Graph Models for Learning and Recognition (GMLR)	2022
Organized within the 37th ACM Symposium on Applied Computing, Brno (Czech Republic)	
